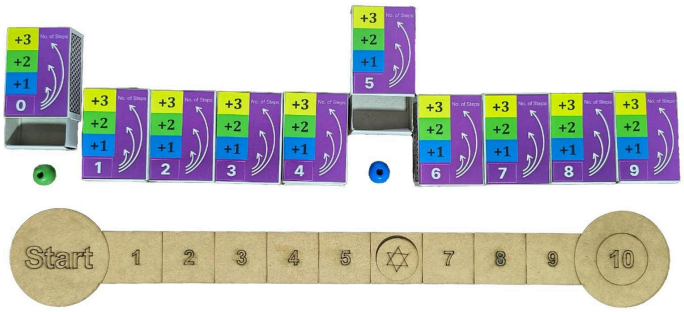
Project 1: Matchboxes that can Learn a Game!!!

Grade 6



Overview

How does one learn to play a board game? We watch others playing the games, their moves, get familiar with the rules, and also play ourselves. The more we play the game, the better we get at the game. In a similar way, a computer also learns to play the games. Such learning methods are used very commonly in most AI applications. The project lets you create and train a model made out of matchboxes !!

Learning Outcomes

- *Introduction to AI methods.* The game learning requires applying AI methods to train the model.
- *Game Strategy.* Simplicity in the game, yet non-trivial to figure out the winning strategy, provides a platform for an experiential approach to teaching/learning.
- *Visualising Learning Graph.* The graph of the number of plays and winning by the learning AI model provides insight into the training process.

Material Required (for a group of 3/4 students)

- Print of this A4 sheet template
- Matchboxes and Glue
- Coloured beads (3 colours, 10 each) or dal grains with different colours

Project Plan

Project Activity	Time	Description
Launch the Game	10-15 mins	The teacher will introduce the game and related rules. Draws the 10 boxes with 1 to 10 numbers for the game template. Demonstrate the game by playing it with students for a couple of rounds to let them get clarity about the rules. Template: https://tinyurl.com/nimtemplate
Play & Explore	20-25 mins	The students will form groups of 3 or 4 and play the game among themselves. The teacher suggests a few explorations while letting students play.
Designing an AI Model	1 week	The students will design a training model to play the game using matchboxes.

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Visualisation and Extension	1 week	Visualise the learning curve by plotting the graph: number of games played vs wins. Extend the matchbox based AI model to another game - Don't say 13 !!

The Game

It's a 2-player simple game. The board has positions numbered 1 to 10. The player who reaches the token to the final position (10) wins the game. Players take turns to move forward. On each turn, the token can move 1, 2, or 3 steps forward.



Figure: Game - Reach the 10 first !!

Play and Explore

1. When the opponent places the token at 5, where should you place the token to win?

0	1	2	3	4	5	6	7	8	9	10

2. When the opponent places the token at 4, where should you place the token to win?

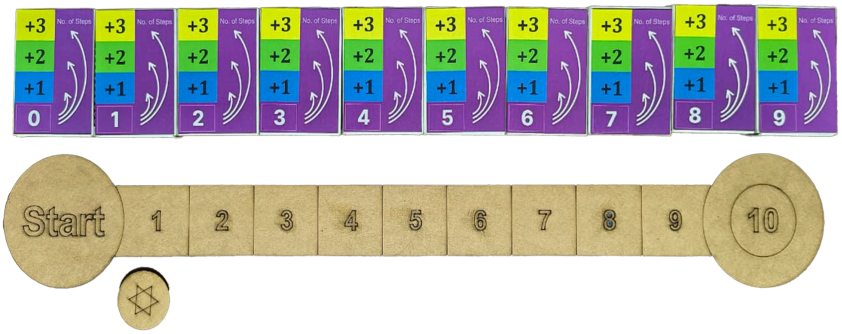
0	1	2	3	4	5	6	7	8	9	10

3. If you were supposed to make the starting move from position 0, how much would you like to move? Why?
4.

Designing an AI Model

Now one player of the game will be a matchbox. Each numbered position on the board will have one matchbox associated with it. Inside every matchbox are three coloured beads (or three coloured grains of dal), each representing a possible action:

- 🔵 Blue bead : +1 : Move forward 1 step
- 🟢 Green bead : +2 : Move forward 2 steps
- 🟡 Yellow bead : +3 : Move forward 3 steps



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Playing Against the Matchboxes
Whenever it is the matchbox machine's turn:

1. Take the matchbox corresponding to the current position.
2. Shake the matchbox, open the drawer, pick a random bead without looking, and note its colour.
3. The colour of the bead tells the machine how many steps to move:
    - Blue → move 1 step
    - Green → move 2 steps
    - Yellow → move 3 steps
4. The bead is kept aside until the game ends.

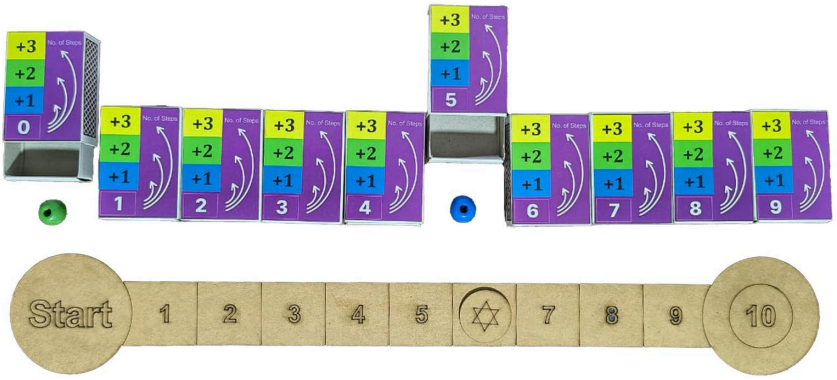
At the beginning, the AI model has no knowledge of good or bad moves, so its actions are mostly random. An example round of the game is shown to understand how the AI model and the student take turns.

1. The AI model plays first. It draws a bead from the matchbox at the starting position. A green bead is drawn, which means the machine advances 2 steps. The token moves to position 2.



2. The student takes the next turn and decides to advance the token 3 steps, placing the token at position 5.

3. Now it is the AI machine's turn again. It draws a bead from the matchbox for position 5. This time, a blue bead is drawn, indicating that the machine should advance 1 step to position 6.



4. The student then advances 1 step, placing the token at position 7.

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5. The AI machine draws a bead from the matchbox for position 7. A yellow bead representing 3 steps. The machine moves the token from 7 to 10.



Since the AI machine reaches the final position (10), it wins the game.

Training the AI Model

After each game finishes, we update the matchboxes depending on whether the AI model won or lost.
If the AI model wins :

- This means the moves it chose were good.
- All beads used in that game are returned to their matchboxes.

This acts as a reward, keeping those actions available for future games.
If the AI model loses :

- The final move made by the matchboxes was a bad decision.
- The bead corresponding to the matchbox's last move is removed.

This acts as a punishment, preventing that poor choice from happening again.
As more games are played:

- Bad moves gradually disappear.
- Good moves remain.
- The machine becomes better at winning.

Visualisation and Extension

While training the matchboxes-based AI model, also plot a graph of games played vs wins. Also, examine whether this graph looks the same or changes when you change the reward rule: now reward winning moves by adding extra beads to those moves.

Extend the algorithm for the **Don't *Say 13*** *game*. A two-player game in which one can move the token up to 2 spaces, and the one who reaches 13 will lose!!

Extend the algorithm for ***Don't Place at 20*** *game*. A two-player game in which one can move the token up to 3 spaces, and the one who reaches 20 will lose!

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